Practical 07: LinearLayout: Horizontal and Vertical

Source: Android Programming Practical(1).pdf | Source pages: 54–56

Aim

To develop and test an Android application demonstrating linearlayout: horizontal and vertical.

Steps to Perform the Practical

1. Create an Android Studio project using Empty Views Activity.

2. Open activity_main.xml and use LinearLayout as shown in the source practical.

3. Set android:orientation to vertical or horizontal as required and add the supplied UI components.

4. Check width, height, gravity, padding and margins for each component.

5. Run the app on an emulator or device.

6. Confirm that the components are arranged horizontally or vertically as intended. The uploaded source ends partway through this practical, so the PDF contains only the pages available in that file.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

III. Basic Layout Management

7. Develop an Android application using LinearLayout (Horizontal and Vertical) to design structured UI.

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="20dp">
   <!-- Title -->
   <TextView
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Student Registration" android:textSize="26sp"
       android:textStyle="bold"
       android:layout_gravity="center" android:layout_marginBottom="20dp"/>
   <!--Vertical Layout Components -->
   <EditText
       android:id="@+id/etName"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:hint="Enter Name"
       android:layout_marginBottom="10dp"/>
```

```
<EditText

    android:id="@+id/etEmail"

    android:layout_width="match_parent"

    android:layout_height="wrap_content"

    android:hint="Enter Email"

    android:layout_marginBottom="20dp"/>


<!-- Horizontal LinearLayout -->

<LinearLayout

    android:layout_width="match_parent"

    android:layout_height="wrap_content"

    android:orientation="horizontal"

    android:gravity="center">


    <Button

        android:id="@+id/btnSubmit"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Submit"/>

    <Button

        android:id="@+id/btnReset"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Reset"

        android:layout_marginStart="20dp"/>

</LinearLayout>
```

```
</LinearLayout>
```